CHAPTER 7

Coordinate geometry: lines, projections and circles

Chapter path

Objectives. Use a normal vector; calculate a projection; determine or recognise a circle; translate a configuration into equations.
Prerequisites. Vector coordinates, scalar product, linear systems, completing the square. The frame is always orthonormal.
Yearly reference point. 5, 7 and 12 January: 6 hours

Diagnostic: six questions, 10 minutes

Answer without consulting the lesson. Mark the questions for which help is needed. This diagnostic is not graded.

1. $(2;-1)\cdot(1;2)$?
2. Does the point $(1;2)$ satisfy $2x-y=0$?
3. Solve $x+y=5$, $x-y=1$.
4. Distance from $(1;2)$ to $(4;6)$?
5. Complete x^2-4x.
6. Solve $x^2+y^2=0$ in $\mathbb{R}^2$.

After the diagnostic. Revisit the flagged topics with the teacher; redo a similar question, then explain the rule used.

Speaking and logic. Present the two conditions that characterise a projection. Give a geometric proof and an algebraic proof of a perpendicular bisector.

Investigate before formalising

Starting from $P(4;3)$ and $d : y = 2x$, compare the distances to three points of d, then find and prove which is the shortest.

Organisation. Individual investigation 5 min, comparison in pairs 5 min, class discussion 5 min. Keep a record of one attempt, even an unfinished one. A conjecture must then be justified.

7.1 Normal vector and Cartesian equation

Definition and results

A nonzero normal vector to a line is orthogonal to every direction vector of that line. If $\vec{n} = (a;b) \neq (0;0)$ is normal and $A(x_A;y_A)$ belongs to d, then d has equation $a(x-x_A)+b(y-y_A) = 0$, that is $ax + by + c = 0$. A possible direction vector is $(-b;a)$.

Status and justification. $M \in d$ is equivalent to $\overrightarrow{AM}\cdot\vec{n} = 0$. In an orthonormal frame, expanding this product gives the equation. Conversely, the set so defined is the line through A orthogonal to n.

Worked example. Through $A(1;2)$ with normal $(2;-3)$: $2(x-1)-3(y-2) = 0$, so $2x-3y+4 = 0$. The vector $(3;2)$ directs the line: product $6-6 = 0$.

Watch out

$(a;b)$ is normal, not a direction vector in general. $(a;b)$ must not be zero.

Immediate application. Equation of the line through $(2;-1)$ with normal $(3;4)$.

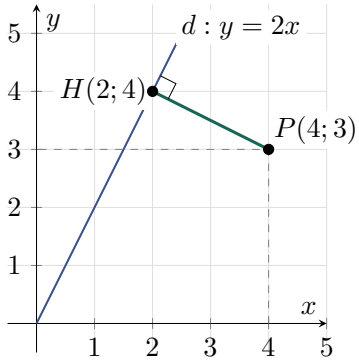
7.2 Orthogonal projection and distance

Definition and results

The projection H of P onto d is characterised by $H \in d$ and $\overrightarrow{PH}$ collinear with a normal vector of d. For $d : ax + by + c = 0$, write $H = P - \lambda(a;b)$, with $\lambda = \dfrac{ax_P + by_P + c}{a^2 + b^2}$. The distance is $PH = \dfrac{|ax_P + by_P + c|}{\sqrt{a^2 + b^2}}$. The distance formula is a deduced supplement, not an isolated routine.

Status and justification. Substitute $x_H = x_P - \lambda a$, $y_H = y_P - \lambda b$ in the equation: $ax_P + by_P + c - \lambda(a^2 + b^2) = 0$. The denominator is positive; we obtain H, then $PH = |\lambda|\sqrt{a^2 + b^2}$.

Worked example. For $P(3;4)$ and $d : x + y - 2 = 0$, $\lambda = \dfrac{3+4-2}{2} = \dfrac{5}{2}$, so $H(\dfrac{1}{2};\dfrac{3}{2})$ and $PH = \dfrac{5}{\sqrt{2}} = \dfrac{5\sqrt{2}}{2}$. Check $x_H + y_H = 2$.



The segment PH is perpendicular to d. Both axes have the same scale; the coordinates of P and H can be read directly.

Watch out

The distance is positive or zero: keep the absolute value. Two checks, membership and perpendicularity, secure the projection.

Immediate application. Project $P(4;1)$ onto $d : x = 2$.

7.3 Equation of a circle

Definition and results

The circle with centre $I(\alpha;\beta)$ and radius $r > 0$ has equation $(x-\alpha)^2 + (y-\beta)^2 = r^2$. An equation $x^2 + y^2 + ux + vy + w = 0$ is handled by completing both squares. If the squared radius obtained is negative, the set is empty; if it is zero, it is a point.

Status and justification. M belongs to the circle if and only if $IM = r$. Since both sides are positive or zero, we can square and use the distance formula in an orthonormal frame.

Worked example. $x^2 + y^2 - 4x + 6y + 9 = 0$ becomes $(x-2)^2 + (y+3)^2 = 4$. Centre $(2;-3)$, radius 2. The signs of the coordinates of the centre are opposite to those appearing in the brackets.

Watch out

A quadratic equation in x and y is not always a circle: check the coefficients and the real radius.

Immediate application. Recognise $x^2 + y^2 + 2x - 8y + 17 = 0$.

7.4 Solving a configuration

Definition and results

An intersection is obtained by solving the equations simultaneously. For a circle and a line parallel to an axis, substituting the constant coordinate gives a quadratic equation. For a perpendicular bisector, write the equality of the squared distances; the quadratic terms cancel.

Status and justification. The equivalences rest on membership of both sets. The equality $MA = MB$ is equivalent to $MA^2 = MB^2$, since the distances are positive.

Worked example. The circle $x^2 + y^2 = 25$ and the line $x = 3$ meet at $(3; 4)$ and $(3; -4)$. The perpendicular bisector of $A(0; 0)$ and $B(4; 2)$ satisfies $x^2 + y^2 = (x - 4)^2 + (y - 2)^2$, so $2x + y = 5$.

Watch out

An intersection is not the union of the solution sets. Check the points in both equations.

Immediate application. Intersect $(x - 1)^2 + (y + 2)^2 = 9$ with $y = 1$.

Exercise path

Start with exercises 1 and 2. Then choose one of exercises 3 or 4 and do it without help. Exercises 5 to 7 are a reserve for transfer, investigation or programming: the teacher designates one. Exercise 8 is revisited after a delay.

Exercise 7.1 – Constructing a line

The plane is equipped with an orthonormal frame. The line d passes through $A(-1; 3)$ and has the vector $\vec{n} = (2; 5)$ as a normal vector.

1. Write the condition for a point $M(x; y)$ to belong to d using a scalar product. Deduce a Cartesian equation of d.
2. Give a nonzero direction vector of d, then the coordinates of a point of d other than A.
3. Check your point by substitution in the equation.

Exercise 7.2 – Calculating a projection

In an orthonormal frame, given the point $P(4; 3)$ and the line d with equation $2x - y = 0$. Let H be the orthogonal projection of P onto d.

1. Determine the coordinates of H by translating the two conditions: H belongs to d and $\overrightarrow{PH}$ is normal to d.
2. Check these two conditions separately with the coordinates obtained.
3. Calculate the distance from P to the line d in exact form.

Exercise 7.3 – Recognising a circle

Consider the set of points $M(x;y)$ of the orthonormal plane whose coordinates satisfy

$$x^2 + y^2 - 6x + 4y - 12 = 0.$$

1. Show that this set is a circle and state its centre and radius.
2. Determine all its points of intersection with the line with equation $y = -2$.
3. Interpret geometrically the position of these two points on the circle.

Exercise 7.4 – Perpendicular bisector

In an orthonormal frame, given $A(-1;1)$ and $B(3;3)$. We look for an equation of the perpendicular bisector of $[AB]$.

1. Determine this equation using the geometric properties of the perpendicular bisector. Explain the choice of a point and of a normal vector.
2. Check that the midpoint of $[AB]$ satisfies the equation obtained.

Exercise 7.5 – The best access

A straight road is represented by the line d with equation $y = 2x$ in an orthonormal frame whose unit of length is 1 km. A building is located at the point $P(4;3)$. We want to build a straight access path joining the building to a point on the road.

1. Determine the point of the road that makes this access as short as possible. Calculate the exact length of the access.
2. Justify that your choice is optimal among all the points of the road, by comparing the distances using a right triangle. A single observation on a drawing is not enough.

Exercise 7.6 – Parameter and set

For a real parameter k, define $\mathcal{E}_k$ as the set of points $M(x;y)$ whose coordinates satisfy

$$x^2 + y^2 - 2x + 4y + k = 0.$$

1. Determine the nature of $\mathcal{E}_k$ according to the value of k. Organise the answer in disjoint cases.
2. When this set is a circle, state its centre and radius; in the other cases, describe exactly the set obtained.

Exercise 7.7 – A parabola as a locus

Optional extension. In an orthonormal frame, fix the point $F(0;2)$. We look for the points $M(x;y)$ whose distance to F is equal to the distance to the x-axis.

1. Express the square of each of the two distances in terms of x and y.
2. Deduce an equation of the set sought. Check the converse in order to justify that this equation describes exactly all the points sought.

Exercise 7.8 – Delayed review

In an orthonormal frame, $\mathcal{C}$ is the circle with diameter $[AB]$, with $A(0;0)$ and $B(4;0)$. Consider the line d with equation $x = 5$.

1. Determine whether $\mathcal{C}$ and d have a common point, by solving an equation.
2. Confirm the answer by comparing the distance from the centre of the circle to the line with the radius of the circle.

Homework. Do only the task designated at the end of the session, in the dated progression: 10–20 min at most. The exercises depend on the topics already studied, so some questions are deferred. Revisit exercise 8 one week after the chapter (10 min). If the allotted time is exceeded, note the difficulty and bring your attempts; do not pile up the whole reserve.

To do in the CQFD textbook

These exercises are done in your textbook. The progression indicates the session; they replace the homework planned for that day (10 to 20 minutes in total).

CQFD, exercise 18 p. 279, questions a, b, c and d. Read a nonzero normal vector for each equation, without solving for y.

CQFD, exercise 21 p. 279, questions a and b. Write the condition of orthogonality to the normal, then check that the given point belongs to the line.

MCQ 1: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. A normal to $2x - 3y + 1 = 0$ is
   A. $(3; 2)$
   B. $(2; -3)$
   C. $(2; 3)$
   D. $(1; 0)$
2. A direction vector of $x + 2y = 0$ is
   A. $(-2; 1)$
   B. $(1; 2)$
   C. $(2; 1)$
   D. $(0; 2)$
3. Centre of $(x + 2)^2 + (y - 3)^2 = 4$:
   A. $(2; 3)$
   B. $(-2; -3)$
   C. $(3; -2)$
   D. $(-2; 3)$
4. Radius of $x^2 + y^2 = 9$:
   A. 9
   B. 81
   C. 3
   D. $\sqrt{3}$
5. $x^2 + y^2 = -1$ describes
   A. a circle
   B. the empty set
   C. the origin
   D. a line

MCQ 2: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. Projection of $(3; 4)$ onto the x-axis:
   A. $(0; 4)$
   B. $(0; 0)$
   C. $(3; 0)$
   D. $(4; 3)$
2. Distance from $(1; 5)$ to $y = 2$:
   A. 3
   B. 5
   C. 1

D. $\sqrt{10}$

3. For $ax + by + c = 0$ to define a line, we need
   A. $a \neq 0$ alone
   B. $b \neq 0$ alone
   C. $c \neq 0$
   D. $(a; b) \neq (0; 0)$
4. $x^2 + y^2 = 0$ describes
   A. a circle of radius 1
   B. the single point O
   C. a line
   D. the plane
5. Circle $x^2 + y^2 = 4$ and line $x = 2$: number of common points
   A. 0
   B. 2
   C. 1
   D. infinitely many

True or false? Justify.

1. The orthogonal projection of P onto a line is unique.
2. Every equation $x^2 + y^2 + c = 0$ describes a circle of positive radius.
3. A line can meet a circle at three distinct points.

Self-assessment

I give a nonzero normal; I check the projection; I complete both squares; I distinguish circle, point and empty set.
For each objective, indicate: achieved without help; achieved with help; to be reworked. Write one example and one mistake to avoid.

Solutions and teaching notes for chapter 7

Teacher use. The answers below concern the tasks not worked in the lesson. First have students state their method; give graded help before the solution. The durations are estimates to be calibrated in class.

Diagnostic and remediation

1. 0. *Action:* Revisit the orthogonality criterion.
2. Yes: $2 - 2 = 0$. *Action:* Substitute x and y simultaneously.
3. $x = 3$, $y = 2$. *Action:* Add, then substitute.
4. 5. *Action:* Calculate the vector, then its norm.
5. $(x-2)^2 - 4$. *Action:* Expand to check.
6. The single pair $(0;0)$. *Action:* Use the sum of two non-negative squares.

Investigation activity

The trials suggest H(2;4), but Pythagoras in PHM proves minimality. The projection is calculated with the normal (2;-1).

Immediate applications

1. $3(x-2) + 4(y+1) = 0$, that is $3x + 4y - 2 = 0$.
2. $H(2;1)$ and distance 2; the normal is horizontal.
3. $(x+1)^2 + (y-4)^2 = 0$: the single point $(-1;4)$, not a circle of strictly positive radius.
4. $(x-1)^2 = 0$, so the only point is $(1;1)$; the line is tangent to the circle.

Exercises: method and results

Exercise 7.1 – Constructing a line. $2(x+1) + 5(y-3) = 0$, that is $2x + 5y - 13 = 0$. Direction $(5;-2)$, so $B(4;1)$ is on the line; check $8 + 5 - 13 = 0$.

Exercise 7.2 – Calculating a projection. Normal $(2;-1)$; $\lambda = \dfrac{8-3}{5} = 1$. $H = (4;3) - (2;-1) = (2;4)$. $2(2) - 4 = 0$ and $\overrightarrow{PH} = (-2;1)$ is normal; distance $\sqrt{5}$.

Exercise 7.3 – Recognising a circle. $(x-3)^2 + (y+2)^2 = 25$; centre $(3;-2)$, radius 5. For $y = -2$, $x = 3 \pm 5$: $(-2;-2)$ and $(8;-2)$.

Exercise 7.4 – Perpendicular bisector. Midpoint I(1;2), normal $\overrightarrow{AB} = (4;2)$. $4(x-1) + 2(y-2) = 0$, that is $2x + y - 4 = 0$. I satisfies $2 + 2 - 4 = 0$.

Exercise 7.5 – The best access. The projection H(2;4) minimises PH, of length $\sqrt{5}$ km. For every M on the road, the triangle PHM is right-angled at H and $PM^2 = PH^2 + HM^2 \geq PH^2$. Equality for M=H.

Exercise 7.6 – Parameter and set. $(x-1)^2+(y+2)^2=5-k$. If $k<5$, circle with centre $(1;-2)$ and radius $\sqrt{5-k}$; if $k=5$, a point; if $k>5$, empty.

Exercise 7.7 – A parabola as a locus. $MF^2=x^2+(y-2)^2$, squared distance to the axis y^2. Equality: $x^2-4y+4=0$, that is $y=\dfrac{x^2}{4}+1$. The distances being positive allows the converse.

Exercise 7.8 – Delayed review. Centre $(2;0)$, radius 2; $(x-2)^2+y^2=4$. For x=5, $y^2=-5$, impossible. Interpretation: the line is at distance 3 from the centre, greater than the radius.

CQFD, exercise 18 p. 279, questions a, b, c and d. Possible answers are $(-1;2)$, $(3;-6)$, $(\frac{1}{4};-1)$ and $(1;0)$. Any nonzero multiple works. For $x=4$, write $x-4=0$.

CQFD, exercise 21 p. 279, questions a and b. a: $x-(y-2)=0$, that is $x-y+2=0$. b: $4(x+1)+6(y-3)=0$, that is $2x+3y-7=0$. Substituting the coordinates of the points confirms membership.

MCQ: answers and reasons

MCQ 1.

1. B. Coefficients of x and y.
2. A. Product with the normal is zero.
3. D. Read the coordinates opposite to the signs.
4. C. Positive root of 9.
5. B. Sum of squares is non-negative.

MCQ 2.

1. C. Same x-coordinate, y-coordinate zero.
2. A. Absolute vertical gap.
3. D. Nonzero normal.
4. B. Both squares must be zero.
5. C. $y^2=0$.

True or false

1. True: through P there is a unique perpendicular to the line; if P belongs to the line, H=P.
2. False: if c=0, a point; if c>0, the empty set.
3. False: substituting a parametrisation of a line into the circle equation gives a quadratic equation with strictly positive leading coefficient, hence at most two roots.

Differentiated teaching. For a technical block, offer only the first step (domain, factor or reference sample space), then withdraw the help. For an independent student, ask for a second method and the treatment of a limiting case. Orally, allow an initial reformulation in the language the student has mastered, then require a precise mathematical sentence in French.